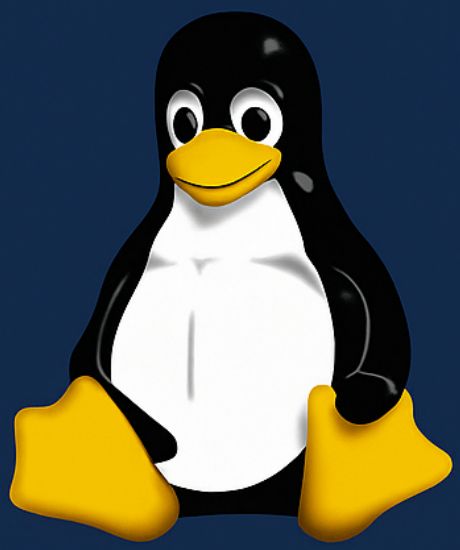
Linux Kernel and Driver Interview Questions



Prepared by

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Process Management

1. How to manipulate the current process states?
2. What are kernel threads?
3. How threads are implemented in Linux kernel?
4. What are different states of a process in Linux?
5. What is difference between process and thread?
6. Generally what resources are shared between threads?
7. What is process descriptor?
8. What is task_struct?
9. What is thread_info structure for?
10. What was the need of thread_info structure?
11. Difference between fork() and vfork()?
12. What is process context?
13. What is zombie process?
14. How parentless process is handled in Linux?

Process Scheduling

1. What is process scheduling?
2. What is cooperative multitasking and pre-emptive multitasking?
3. What is yielding?
4. What is limitation of cooperative multitasking?
5. I/O bound versus Processor bound process?

6. What is process priority?
7. What kind of priority is maintained in Linux?
8. What is nice value?
9. What is virtual run time?
10. What are the available scheduling classes in Linux?
11. Which type of scheduling used in Linux?
12. How next task is picked for scheduling?
13. What is scheduler entry point in Linux?
14. What is waitqueue?
15. How context switching is handled in Linux?
16. What is user preemption and kernel preemption?

Syscalls

1. What is syscalls?
2. How system calls are implemented in Linux?
3. What happens when process in userspace calls a syscall?
4. What is the need of verifying parameter in definition of syscall?
5. What is system call context?
6. Why it is not recommended to write new syscalls?

Interrupt Handling

1. What is interrupt?
2. What is interrupt handler or ISR?
3. What is top halves and bottom halves?

4. How interrupt is registered?
5. What are different interrupt handler flags?
6. How interrupts are freed?
7. What are the considerations to take care while writing interrupt handler?
8. What is interrupt context?
9. How to disable and enable interrupts?
10. What are different bottom halves techniques in Linux?
11. What is tasklets, softirq and workqueues and difference among them?
12. When to choose which bottom halves?
13. How to implement softirq, tasklets and workqueues?
14. How to schedule tasklet?
15. What is ksoftirqd?
16. How to disable bottom halves?
17. How locking between bottom halves is handled?

Kernel Synchronization

1. What is synchronization?
2. What is critical section?
3. What is race condition?
4. Why we need to take care of synchronization?
5. What are various synchronization techniques in Linux?
6. What is deadlock?
7. What is atomic operation?

8. What is spinlock?
9. What is reader-writer spin lock?
10. What is semaphore?
11. What is binary semaphore?
12. What is difference between semaphore and spinlock?
13. When to choose between spinlock and semaphore?
14. What is difference between semaphore and mutex?
15. What is preemption disabling and what is the use of this?

Memory Management

1. How memory is managed in Linux?
2. What are pages?
3. What are different memory zones in Linux?
4. How to allocate pages?
5. How to free pages?
6. What is kmalloc and what are action modifiers?
7. What is zone modifier in Linux?
8. What is vmalloc?

Virtual File System

1. What is virtual file system and what is the need of it in Linux?
2. What are different object types in VFS?
3. What are the operations possible on inode and superblock objects?

Process Address Space

1. What is process address space?
2. What is memory descriptor in Linux and which structure represents it?
3. How to allocate and destroy a memory descriptor?
4. Does kernel thread have any association with mm_struct?
5. What is VMA and what are various VMA operations possible in Linux kernel?
6. How to manipulate memory area in Linux?
7. What is mmap and do_mmap()?
8. What is page table?

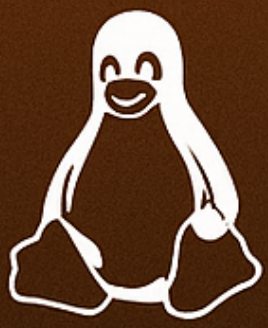
Device Driver Questions

1. What is device driver and what is the need of it?
2. What are different kinds of devices?
3. What is module in Linux?
4. How modules are loaded in Linux?
5. Difference between insmod and modprobe?
6. How parameters are shared between driver modules?
7. What are IOCTLs?
8. What is syscalls?
9. What are the benefits of syscalls?
10. How character driver is registered in Linux?
11. What is init and exit function of a driver?

12. How and when init and exit function of driver get called?
13. What is probe function?
14. When probe is called?
15. What is platform device?
16. What is device tree?
17. What are the benefits of device tree over board files?
18. What is sysfs and procfs?
19. How logs are printed in Linux kernel and what are the log levels?
20. What is copy_to_user and copy_from_user?
21. What do you mean by kernel configuration and what are the ways of configuring kernel?
22. What is menuconfig?
23. What is ioremap?
24. What is segmentation fault?
25. What are the various ways of debugging Linux kernel?
26. How Linux kernel boots?
27. What is zImage and bzImage?
28. What are different boot arguments in Linux?
29. How parameters are passed from boot loader to kernel?
30. What is ATAGs?
31. From which file kernel execution starts?
32. What is bootloader?
33. What is primary and secondary bootloader?
34. Why do we need two bootloaders?

35. Difference between poll and select?
36. What is priority inheritance and priority inversion?
37. What are different types of kernel?
38. What is DMA?
39. What is cache coherency?
40. What is copy on write?
41. What is highmem and lowmem?
42. What happens if we pass invalid address from userspace by using ioctl?
43. What are different IPC mechanisms in Linux?
44. What are sockets?
45. How page fault is handled in Linux?
46. Difference between memory based IO and port based IO?
47. What is I2C and SPI?
48. How physical to virtual translation works in Linux?
49. What is thrashing, segmentation and fragmentation?
50. What is preempt_count and what is the need of it?

Answers of all these questions are part of our Embedded System Interview Prep Bootcamp



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